20261029 Midterm

for (MATH261) Discrete Mathematics

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Problem. Prove that $\sqrt{2}$ is irrational.

Solution. Suppose that $\sqrt{2} = p/q$ for relatively prime positive integers p, q. Then $p^2 = 2q^2$, so p^2 and hence p are even. Write $p = 2r$. Substitution gives $4r^2 = 2q^2$, or $q^2 = 2r^2$, so q is even as well. Thus 2 divides both p and q, contrary to $\gcd(p, q) = 1$.

□

Problem. Prove that there exist irrational real numbers x, y such that x^y is rational.

Solution. Set $a = (\sqrt{2})^{\sqrt{2}}$. If $a \in \mathbb{Q}$, take $x = y = \sqrt{2}$, obtaining $x^y = a \in \mathbb{Q}$. Otherwise a is irrational; take $x = a$ and $y = \sqrt{2}$. Then

$$x^y = \left((\sqrt{2})^{\sqrt{2}}\right)^{\sqrt{2}} = (\sqrt{2})^2 = 2 \in \mathbb{Q}.$$

In either case both x and y are irrational, as required.

□

Problem. Prove that among any $n + 1$ distinct positive integers not exceeding $2n$, there are two such that one divides the other.

Solution. Each integer $m \in [2n]$ has a unique representation $m = 2^k u$, where $k \in \mathbb{N}_0$ and u is odd. There are exactly n possible odd values of u in $[2n]$. By the Pigeonhole Principle, two of the $n + 1$ selected integers have the same odd part, say $2^a u$ and $2^b u$. Assuming $a < b$, we have $2^b u = 2^{b-a}(2^a u)$, so the first divides the second.

□

Problem. Prove that every sequence of n^2+1 distinct real numbers contains a strictly increasing subsequence of length $n + 1$ or a strictly decreasing subsequence of length $n + 1$.

Solution. Write the sequence as $a_1, \ldots, a_{n^2+1}$. For each i, let u_i be the maximum length of a strictly increasing subsequence ending at a_i, and let v_i be the corresponding maximum length for a strictly decreasing subsequence. Suppose neither type of subsequence has length $n + 1$. Then $u_i, v_i \in [n]$ for every i, so there are only n^2 possible pairs (u_i, v_i).

These pairs must, however, be distinct. Indeed, if $i < j$ and $a_i < a_j$, appending a_j to a longest increasing subsequence ending at a_i gives $u_j \geq u_i + 1$. If $a_i > a_j$, the analogous argument gives $v_j \geq v_i + 1$. In either case $(u_i, v_i) \neq (u_j, v_j)$. This produces $n^2 + 1$ distinct pairs in $[n] \times [n]$, a contradiction.

□

Problem. Construct a bijection between $\mathbb{N}$ and $\mathbb{Z}$, and hence prove that $|\mathbb{N}| = |\mathbb{Z}|$.

Solution. Define $f\colon \mathbb{N} \to \mathbb{Z}$ by

$$f(n) = \begin{cases} 0, & n = 1, \\ k, & n = 2k \quad (k \geq 1), \\ -k, & n = 2k+1 \quad (k \geq 1). \end{cases}$$

Every integer has a preimage: $0 = f(1)$, $k = f(2k)$, and $-k = f(2k+1)$ for $k \geq 1$. Thus f is surjective. The three displayed cases have disjoint images (zero, positive integers, and negative integers, respectively), and within each case the rule is injective. Hence f is injective as well, so it is a bijection.

□

Problem. There are $2n$ people waiting in line to buy theater tickets costing 50 cents each. Of these, n have one 50-cent coin each and the other n have one 1 bill each. The cashier initially has no change. How many orders of the line allow every customer to purchase a ticket without a shortage of change? Give the answer both when people with the same form of payment are indistinguishable and when all $2n$ people are distinct.

Solution. Write $+1$ for a customer who pays 50 cents and -1 for one who pays 1. The cashier can serve everyone precisely when each initial partial sum is nonnegative, since a 1-bill customer consumes one 50-cent coin in change. The line therefore corresponds to a Catalan word with n entries of each type. The number of payment-type sequences is

$$C_n = \frac{1}{n+1}\binom{2n}{n}.$$

Thus the answer for indistinguishable people of each type is C_n. If all people are distinct, there are $n!$ assignments of the 50-cent customers and $n!$ assignments of the 1-bill customers to the corresponding positions of every admissible payment-type sequence. The answer is therefore

$$n!^2 C_n = \frac{(2n)!}{n+1}.$$

□

Problem. Let $f_0 = 0$, $f_1 = 1$, and $f_n = f_{n-1} + f_{n-2}$ for $n \geq 2$. Find an explicit formula for f_n.

Solution. The characteristic polynomial is $r^2 - r - 1$, with distinct roots

$$\alpha = \frac{1+\sqrt{5}}{2}, \qquad \beta = \frac{1-\sqrt{5}}{2}.$$

Consequently $f_n = A\alpha^n + B\beta^n$ for constants A, B. The initial conditions yield $A + B = 0$ and $A\alpha + B\beta = 1$. Since $\alpha - \beta = \sqrt{5}$, we obtain $A = 1/\sqrt{5}$ and $B = -1/\sqrt{5}$. Thus

$$f_n = \frac{\alpha^n - \beta^n}{\sqrt{5}} = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right].$$

□

Problem. Solve $a_n = 6a_{n-1} - 9a_{n-2}$ for $n \geq 2$, with $a_0 = 1$ and $a_1 = 6$.

Solution. The characteristic equation is $(r-3)^2 = 0$, so the general solution is $a_n = (A + Bn)3^n$. Since $a_0 = A = 1$ and $a_1 = 3(A + B) = 6$, we find $B = 1$. Hence

$$a_n = (n+1)3^n \qquad (n \geq 0).$$

□

Problem. Solve

$$a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3} \qquad (n \geq 3),$$

where $a_0 = 1$, $a_1 = -2$, and $a_2 = -1$.

Solution. The characteristic polynomial is $r^3 + 3r^2 + 3r + 1 = (r+1)^3$. The root -1 has multiplicity three, so

$$a_n = (A + Bn + Cn^2)(-1)^n.$$

Substitution of the initial conditions gives

$$A = 1, \qquad A + B + C = 2, \qquad A + 2B + 4C = -1.$$

Thus $B + C = 1$ and $B + 2C = -1$, giving $B = 3$ and $C = -2$. Therefore

$$a_n = (1 + 3n - 2n^2)(-1)^n \qquad (n \geq 0).$$

□

Problem. Find all solutions of $a_n = 3a_{n-1} + 2n$ for $n \geq 2$, and determine the solution satisfying $a_1 = 3$.

Solution. The associated homogeneous recurrence has solutions $A3^n$. As the forcing term is linear in n, seek a particular solution $p_n = cn + d$. Substitution gives

$$cn + d = 3\big(c(n-1) + d\big) + 2n,$$

so $c = 3c + 2$ and $d = -3c + 3d$. Thus $c = -1$ and $d = -\frac{3}{2}$, and every solution is

$$a_n = A3^n - n - \frac{3}{2}.$$

The condition $a_1 = 3$ yields $3 = 3A - 1 - \frac{3}{2}$, hence $A = \frac{11}{6}$. The required solution is

$$a_n = \frac{11}{6}3^n - n - \frac{3}{2}.$$

□

Problem. Find all solutions of

$$a_n = 5a_{n-1} - 6a_{n-2} + 7^n \qquad (n \geq 2).$$

Solution. The characteristic polynomial of the associated homogeneous recurrence is $(r-2)(r-3)$, so its solutions are $A2^n + B3^n$. As 7 is not a characteristic root, try $p_n = C7^n$. Substitution gives

$$C7^n = 5C7^{n-1} - 6C7^{n-2} + 7^n,$$

which, upon division by 7^{n-2}, becomes $49C = 35C - 6C + 49$. Hence $20C = 49$ and $C = \frac{49}{20}$. All solutions are therefore

$$a_n = A2^n + B3^n + \frac{49}{20}7^n.$$

□

Problem. Let $a_0 = 0$ and

$$a_n = \frac{2(n-1)}{n} a_{n-1} + \frac{1}{n} \qquad (n \geq 1).$$

Find an explicit formula for a_n by two different methods.

Solution 1. Set $b_n = n a_n$ for $n \geq 1$. Multiplying the recurrence by n shows that $b_1 = 1$ and

$$b_n = 2b_{n-1} + 1 \qquad (n \geq 2).$$

Thus $b_n + 1 = 2(b_{n-1} + 1)$, whence $b_n + 1 = 2^{n-1}(b_1 + 1) = 2^n$. Therefore

$$a_n = \frac{2^n - 1}{n} \qquad (n \geq 1), \qquad a_0 = 0.$$

□

Solution 2. The first step gives $a_1 = 1$. For $n \geq 2$, the coefficient $2(n-1)/n$ is nonzero, so the first-order variable-coefficient solution formula can be applied starting at $n = 2$:

$$\begin{aligned} a_n &= a_1 \prod_{j=2}^{n} \frac{2(j-1)}{j} + \sum_{k=2}^{n} \frac{1}{k} \prod_{j=k+1}^{n} \frac{2(j-1)}{j} \\ &= \frac{2^{n-1}}{n} + \frac{1}{n} \sum_{k=2}^{n} 2^{n-k} \\ &= \frac{2^{n-1}}{n} + \frac{2^{n-1} - 1}{n} = \frac{2^n - 1}{n}. \end{aligned}$$

Empty products are 1, so the computation also covers $n = 2$. The case $n = 1$ holds directly and $a_0 = 0$ is the given initial value.

□

Problem. Let $m \in \mathbb{N}$. Prove the following identities for $|x| < 1$:

(a) $$(1+x)^{-m} = \sum_{n=0}^{\infty} \binom{m+n-1}{n} (-x)^n.$$

(b) $$(1-x)^{-m} = \sum_{n=0}^{\infty} \binom{m+n-1}{n} x^n.$$

Solution. The generalized binomial coefficient satisfies

$$\binom{-m}{n} = \frac{(-m)(-m-1)\cdots(-m-n+1)}{n!} = (-1)^n \frac{(m+n-1)!}{(m-1)!n!} = (-1)^n \binom{m+n-1}{n}.$$

By the Generalized Binomial Theorem,

$$(1+x)^{-m} = \sum_{n=0}^{\infty} \binom{-m}{n} x^n = \sum_{n=0}^{\infty} \binom{m+n-1}{n} (-x)^n,$$

which proves (a). Replacing x by $-x$ proves (b). Both expansions converge for $|x| < 1$.

□

Problem. Use an ordinary generating function to solve $a_n = 5a_{n-1} - 6a_{n-2}$ for $n \geq 2$, where $a_0 = 0$ and $a_1 = 2$.

Solution. Let $F(x) = \sum_{n\geq 0} a_n x^n$. Multiplying the recurrence by x^n and summing over $n \geq 2$ gives

$$F(x) - 2x = 5x\big(F(x) - a_0\big) - 6x^2 F(x).$$

Since $a_0 = 0$, it follows that

$$F(x) = \frac{2x}{1 - 5x + 6x^2} = \frac{-2}{1-2x} + \frac{2}{1-3x}.$$

The geometric-series identity $(1-cx)^{-1} = \sum_{n\geq 0} c^n x^n$ gives

$$a_n = -2 \cdot 2^n + 2 \cdot 3^n = 2(3^n - 2^n) \qquad (n \geq 0).$$

□

Problem. Use an ordinary generating function to solve $a_n = -a_{n-1} + 6a_{n-2}$ for $n \geq 2$, where $a_0 = 1$ and $a_1 = 3$.

Solution. Let $F(x) = \sum_{n\geq 0} a_n x^n$. After multiplying by x^n and summing for $n \geq 2$,

$$F(x) - 1 - 3x = -x\big(F(x) - 1\big) + 6x^2 F(x).$$

Thus

$$F(x) = \frac{1 + 4x}{1 + x - 6x^2} = \frac{6}{5}\frac{1}{1-2x} - \frac{1}{5}\frac{1}{1+3x}.$$

Extracting coefficients yields

$$a_n = \frac{6}{5}\,2^n - \frac{1}{5}(-3)^n \qquad (n \geq 0).$$

□

Problem. Use an ordinary generating function to solve

$$b_n = b_{n-1} + 2b_{n-2} + 2^n \qquad (n \geq 2),$$

where $b_0 = 2$ and $b_1 = 1$.

Solution. Set $F(x) = \sum_{n\geq 0} b_n x^n$. As $\sum_{n\geq 2} 2^n x^n = (1-2x)^{-1} - 1 - 2x$, summation of the recurrence gives

$$F(x) - 2 - x = x\big(F(x) - 2\big) + 2x^2 F(x) + \frac{1}{1-2x} - 1 - 2x.$$

Rearranging and simplifying,

$$\begin{aligned} F(x)(1 - x - 2x^2) &= 1 - 3x + \frac{1}{1-2x}, \\ F(x) &= \frac{2 - 5x + 6x^2}{(1-2x)^2(1+x)} \\ &= -\frac{1}{9(1-2x)} + \frac{2}{3(1-2x)^2} + \frac{13}{9(1+x)}. \end{aligned}$$

Using $[x^n](1-2x)^{-2} = (n+1)2^n$, we obtain

$$b_n = -\frac{1}{9}2^n + \frac{2}{3}(n+1)2^n + \frac{13}{9}(-1)^n \qquad (n \geq 0).$$

□

Problem. Let $(C_n)_{n\geq 0}$ denote the Catalan numbers, characterized by $C_0 = 1$ and

$$C_n = \sum_{j=0}^{n-1} C_j C_{n-1-j} \qquad (n \geq 1).$$

(a) Find the ordinary generating function $F(x) = \sum_{n\geq 0} C_n x^n$.

(b) Using the generating function and the Generalized Binomial Theorem, prove that $C_n = \dfrac{1}{n+1}\binom{2n}{n}$.

(c) Set $a_n = nC_n$. Find $G(x) = \sum_{n\geq 0} a_n x^n$.

Solution.

(a) Multiplying the recurrence by x^n and summing over $n \geq 1$ yields

$$F(x) - 1 = x \sum_{m\geq 0} \left(\sum_{j=0}^{m} C_j C_{m-j} \right) x^m = xF(x)^2.$$

Hence $xF(x)^2 - F(x) + 1 = 0$, giving $F(x) = (1 \pm \sqrt{1-4x})/(2x)$. The positive sign has a pole at $x = 0$, whereas $F(0) = 1$; thus

$$\boxed{F(x) = \frac{1 - \sqrt{1-4x}}{2x}.}$$

(b) The Generalized Binomial Theorem gives

$$\sqrt{1-4x} = \sum_{j\geq 0} \binom{1/2}{j} (-4x)^j.$$

Consequently, for $n \geq 0$,

$$\begin{aligned} C_n = [x^n]F(x) &= -\frac{1}{2}\binom{1/2}{n+1}(-4)^{n+1} \\ &= \frac{2^n(2n-1)!!}{(n+1)!} = \frac{(2n)!}{n!(n+1)!} = \frac{1}{n+1}\binom{2n}{n}, \end{aligned}$$

where $(-1)!! = 1$ for $n = 0$.

(c) By formal differentiation,

$$G(x) = \sum_{n\geq 0} nC_n x^n = xF'(x).$$

Writing $s = \sqrt{1-4x}$ and $F(x) = 2/(1+s)$ gives

$$G(x) = \frac{4x}{s(1+s)^2} = \frac{1-s}{s(1+s)} = \frac{1}{\sqrt{1-4x}} - \frac{1-\sqrt{1-4x}}{2x}.$$

□

Problem. Let D_n denote the number of derangements of an n-element set, with $D_0 = 1$ and $D_n = nD_{n-1} + (-1)^n$ for $n \geq 1$. Find the exponential generating function of $(D_n)_{n\geq 0}$.

Solution. Let $F(x) = \sum_{n\geq 0} D_n x^n/n!$. Multiplying the recurrence by $x^n/n!$ and summing for $n \geq 1$ gives

$$\begin{aligned} F(x) - 1 &= \sum_{n\geq 1} D_{n-1}\frac{x^n}{(n-1)!} + \sum_{n\geq 1}(-1)^n\frac{x^n}{n!} \\ &= xF(x) + (e^{-x} - 1). \end{aligned}$$

Therefore $(1-x)F(x) = e^{-x}$ and

$$\boxed{F(x) = \frac{e^{-x}}{1-x}.}$$

□

Problem. Let $B_0 = 1$ and let the Bell numbers satisfy

$$B_{n+1} = \sum_{k=0}^{n}\binom{n}{k}B_k \qquad (n \geq 0).$$

Find the exponential generating function of $(B_n)_{n\geq 0}$.

Solution. Let $G(x) = \sum_{n\geq 0} B_n x^n/n!$. Termwise differentiation and the Bell recurrence give

$$\begin{aligned} G'(x) &= \sum_{n\geq 0} B_{n+1}\frac{x^n}{n!} \\ &= \sum_{n\geq 0}\left(\sum_{k=0}^{n}\binom{n}{k}B_k\right)\frac{x^n}{n!} = e^x G(x), \end{aligned}$$

where the last equality is the exponential generating function product formula. Since $G(0) = 1$, the differential equation implies $\frac{d}{dx}(e^{-e^x}G(x)) = 0$, hence $e^{-e^x}G(x) = e^{-1}$. Thus

$$\boxed{G(x) = e^{e^x - 1}.}$$

□

Problem. For each $k \in \mathbb{N}_0$, let $S(n,k)$ be the Stirling number of the second kind, with $S(0,0) = 1$ and $S(n,0) = 0$ for $n \geq 1$.

(a) For fixed k, find the ordinary generating function $F_k(x) = \sum_{n\geq 0} S(n,k)x^n$.

(b) For fixed k, find the exponential generating function $G_k(x) = \sum_{n\geq 0} S(n,k)x^n/n!$.

Solution.

(a) The recurrence $S(n,k) = kS(n-1,k) + S(n-1,k-1)$ for $n \geq 1$ (with $S(0,k) = 0$ when $k \geq 1$) implies

$$F_k(x) = kxF_k(x) + xF_{k-1}(x).$$

Since $F_0(x) = 1$, iteration yields, for $k \geq 1$,

$$\boxed{F_k(x) = \frac{x^k}{\prod_{j=1}^{k}(1-jx)}.}$$

The empty-product convention makes this valid for $k = 0$ as well.

(b) The surjection-counting formula is

$$k!S(n,k) = \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} j^n.$$

Multiplying by $x^n/n!$ and summing over $n \geq 0$ gives

$$\begin{aligned} G_k(x) &= \frac{1}{k!} \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} \sum_{n \geq 0} \frac{(jx)^n}{n!} \\ &= \frac{1}{k!} \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} e^{jx} = \boxed{\frac{(e^x - 1)^k}{k!}}. \end{aligned}$$

The formula is also valid at $k = 0$, when $G_0(x) = 1$.

□